

MATH 345 Skeleton Notes

Unit 6: Constraints and SVD

Section 6.2: Lagrange multipliers and parallel gradients

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Note: Lagrange multipliers as parallel gradients The Lagrange multiplier condition $\nabla f(a) = \lambda \nabla g(a)$ means that, at a constrained optimum, the objective gradient is parallel to the constraint gradient. Equivalently, along the allowed tangent directions to the constraint, there is no first-order increase or decrease.

This uses four earlier ideas: dot products, orthogonality, level sets, and gradients.

We now discuss the method of Lagrange multipliers, which is a method of finding potential local minima and local maxima of a function on a *hypersurface* S in $\mathbb{R}^n$ given as the level set of a function, i.e. when $S = \{\mathbf{x} \in \mathbb{R}^n : g(\mathbf{x}) = k\}$, for some differentiable scalar-valued function g and some $k \in \mathbb{R}$.

Definition: Critical Points Consider a surface of the form $S = \{\mathbf{x} \in \mathbb{R}^n : g(\mathbf{x}) = k\}$, where g is a differentiable scalar-valued function with $\nabla g(\mathbf{x}) \neq 0$ for all $\mathbf{x} \in S$. If a point $\mathbf{a} \in S$ is in the interior of the domain of a differentiable function f , then we say that $\mathbf{a}$ is a

constrained critical point of f on S if there exists $\lambda \in \mathbb{R}$ such that

$$\nabla f(\mathbf{a}) = \lambda \nabla g(\mathbf{a}).$$

Equivalently, define the

Lagrangian function

$$\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) - \lambda(g(\mathbf{x}) - k).$$

Then $\mathbf{a}$ is a constrained critical point exactly when there exists $\lambda \in \mathbb{R}$ such that

$$\nabla_{\mathbf{x}} \mathcal{L}(\mathbf{a}, \lambda) = \mathbf{0}, \quad \frac{\partial \mathcal{L}}{\partial \lambda}(\mathbf{a}, \lambda) = 0.$$

The *method of Lagrange multipliers* is a way of finding optima of a function f along a hypersurface S given as above. The method consists of finding all critical points of the function f on S (of which there are usually only finitely many). This consists of finding the solutions to the system of vector equations $g(\mathbf{x}) = k$ and $\nabla f(\mathbf{x}) = \lambda \nabla g(\mathbf{x})$. The next theorem shows that any local extremum of f on S must occur at a critical point, so that the method of Lagrange multipliers will find these extrema.

Theorem Consider a surface

$$S = \{\mathbf{x} \in \mathbb{R}^n : g(\mathbf{x}) = k\},$$

and assume ∇g is nonzero on S . If a point $\mathbf{a} \in S$ is a local extremum of a differentiable function f on S , and $\mathbf{a}$ is an interior point of the domain of f , then $\mathbf{a}$ is a constrained critical point of f on S .

In particular, if S is closed and bounded and f is continuous on S , then absolute extrema of f on S exist. Under the same differentiability and nonzero-gradient hypotheses, every absolute extremum of f on S is among the constrained critical points.

Proof The idea is quite simple when sketched geometrically. In order for $\mathbf{a}$ to be a local extremum for f on S , it must be true that for any unit vector $\mathbf{v}$ in the tangent plane to S at a point $\mathbf{a}$, the directional derivative $D_{\mathbf{v}} f(\mathbf{a})$ must be equal to 0 (or else travelling along S in the direction

of $\mathbf{v}$ or $-\mathbf{v}$ will lead the function to increase or decrease, which contradicts the fact that $\mathbf{a}$ is a local extrema). Recalling Definition def-10-1-directional-derivative, it follows that

$$\nabla f(\mathbf{a}) \cdot \mathbf{v} = 0$$

for any vector $\mathbf{v}$ in the tangent plane to S at $\mathbf{a}$, which can *only* be true if $\nabla f(\mathbf{a})$ is normal to the tangent plane to the surface S at $\mathbf{a}$. When S is equal to the level set of a function g , a normal vector to the tangent plane at $\mathbf{a}$ is given by $\nabla g(\mathbf{a})$. So if $\mathbf{a}$ is a local extremum of f on S , then $\nabla f(\mathbf{a})$ and $\nabla g(\mathbf{a})$ must be multiples of one another, i.e. it must be true that

$$\nabla f(\mathbf{a}) = \lambda \nabla g(\mathbf{a})$$

for some $\lambda \in \mathbb{R}$, i.e. so that $\mathbf{a}$ is a constrained critical point of f on S . The final statement follows from the extreme value theorem: if S is closed and bounded and f is continuous on S , then absolute extrema exist, and any absolute extremum is also a local extremum on S . The first part of the theorem then shows that such a point must be a constrained critical point.

Activity

Find the absolute extrema of $f(x, y) = 3x + y$ subject to the constraint $x^2 + y^2 = 10$.

Activity

Find the absolute extrema of the function $f(x, y, z) = x^2 + 2y^2 + 3z^2$, subject to the constraint $x^2 + y^2 + z^2 = 1$.